Evacuation of pedestrians from a single room by using snowdrift game theories

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Game theory is introduced to simulate the complicated interaction relations among the conflicting pedestrians in a pedestrian flow system, which is defined on a square lattice with the parallel update rule. Modified on the traditional lattice gas model, each pedestrian can move to not only an empty site, but also an occupied site. It is found that each individual chooses its neighbor randomly and occupies the site with the probability $W(x \to y) = \frac{1}{1+\exp[-(P_x - U_x)/\kappa]}$, where P_x is the x's payoff representing his personal energy, and U_x is the average payoff of its neighborhood indicating the potential well energy if he stays. Two types of pedestrians are considered, and they interact with their neighbors following the payoff matrix of snowdrift game theory. The cost-to-benefit ratio $r = c/(2b - c)$ (where b is the perfect payoff and c is the labor cost) represents the fear index of the pedestrians in this model. It is found that there exists a moderate value of r leading to the shortest escape time, and the situation for large values of r is better than that for small ones in general. In addition, the pedestrian flow system always arrives at a consistent state in which the two types of walkers have the same number and evolve by the same law irrespectively of the parameters, which can be interpreted as the self-organization effect of pedestrian flow. It is also proven that the time point of the onset of the steady state is unrelated to the scale of the pedestrians and the square lattice. Meanwhile, the system exhibits different dynamics before reaching the consistent state: the number of the two types of walkers oscillates when $P_C > 0.5$ (i.e., probability to change the present strategy), while no oscillation happens for $P_C \leqslant 0.5$. Finally, it is shown that a smaller density of pedestrians ρ induces a shorter average escape time.

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I. INTRODUCTION

Pedestrian flow is a kind of many-body system of strongly interacting persons, which has attracted much attention in recent years [1–14]. In order to investigate the dynamics of pedestrian flow, many models have been developed as well as the experimental studies [15–27]. Henderson has conjectured that the behaviors of pedestrian crowds are similar to gases or fluids [28]. Helbing has used the active walker model to describe human trail formation, and has shown that the pedestrian flow system exhibits various collective phenomena interpreted as self-organization effects [22]. Fukui and Ishibashi have described the movements of pedestrians by means of the cellular automaton model [29].

Pedestrian evacuation dynamics [20,30–35] has been extensively studied in order to determine how to avoid injuries and deaths in the case of emergencies, and the lattice gas model [17,36] is one of the widely used models to simulate the movements of pedestrians. In the lattice gas model, N pedestrians are set on the lattice with $L \times W$, and one site only holds one walker. Each walker moves to the preferential direction performing a biased random walk with no back steps, and can only move to the empty sites. Overlapping on one site is prohibited.

In real situations of an evacuation process based on experimental observations, there are many common features with driven granular media; other phenomena such as "freezing by heating" are even more surprising. Actually, a real pedestrian flow system is neither a continuous model nor a lattice gas model. People are active walkers and their motions are influenced by both subjective reasons (e.g., fear emotions, morals, personal abilities) and objective aspects (e.g., location and number of exits, obstacles). An important task related to the evacuation simulation is how to simulate the effects of subjective aspects and the complicated interaction force among the pedestrians [27,37–44].

Based on this consideration, the snowdrift game theory model [45–52] is proposed for the pedestrian evacuation process. There are three main features of this model. First, the payoff matrix reflects the complicated interaction force among the pedestrians, and the average payoff for each walker indicates the personal energy; meanwhile, the average payoff of its neighborhood represents the potential well energy constraining the walker. Second, the motion rule proposed in this work enriches the dynamics of the system: Although people will select the sites forward, there exist three different motion forms, namely, moving forward, staying on the original site, and being pushed onto the site backwards. Furthermore, we introduce a parameter P_C (i.e., the probability to change the present strategy) to simulate the fear emotion and the personal contingency ability. Researching this model, we found that the average escape time decreases as the density of pedestrians decreases, and a consistent state is reached regardless of the parameters, which can be interpreted as the self-organization effects reported in the previous work.

The paper is organized as follows. In Sec. II, a description of the model is proposed in detail. Numerical simulations and the corresponding analysis are presented in Sec. III. Conclusions are drawn in Sec. IV.

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TABLE I. The payoff matrix of the snowdrift game.

	C	D	E
C	$(R\ R)$	$(S\ T)$	$(1-)$
D	$(T\ S)$	$(P\ P)$	$(1-)$
E	(-1)	(-1)	$(--)$

II. MODELS

The evacuation of pedestrians from a single room with one door downward is studied on a square lattice with $L \times W$, where W is the width of the house and L is the length. N pedestrians are set on the lattice and each site can only hold a single walker. The density of pedestrian flow ρ is defined as $\rho = N/(L \times W)$. Each pedestrian is able to walk toward the position of the exit with no back steps, and can move along the wall but never go out of the wall. The walkers are deleted on the lattice once they arrive at the exit. There are two types of pedestrians (cooperators C and defectors D) considered, and the snowdrift game model is introduced indicating the complicated interaction among the conflict individuals: Each walker interacts with all its neighbors and gets the payoff following the payoff matrix in Table I. To be specific, for mutual cooperation, each walker receives the rewards ($R = b - c/2$), two defectors receive punishment ($P = 0$), while a cooperator and a defector gets the sucker's payoff ($S = b - c$) and the temptation ($T = b$), respectively. Here, b is the perfect benefit and c is the labor cost in the game. Without losing generality, $b - c/2$ is usually set to be 1 so that the behavior of the snowdrift game can be investigated by a single parameter r, $r = c/2 = c/(2b - c)$. Usually, $T = 1 + r$, $R = 1$, $S = 1 - r$, and $P = 0$, where $0 < r < 1$. Meanwhile, a cooperator and defector receive 1 when there is an empty (E) site in its neighborhood. The payoff P_x of pedestrian x is the average of the accumulated payoff from all of its neighbors representing the personal energy.

At each step, each pedestrian x will choose a neighbor site y randomly, and occupy the site with the probability $W(x \to y)$,

$$W(x \to y) = \frac{1}{1 + \exp[-(P_x - U_x)/\kappa]}, \tag{1}$$

where U_x is the average payoff of its neighborhood indicating that the potential well energy x stays in, and U_x is increased by 1 if he is on the boundary by a wall (the potential well energy increases). Here, κ denotes the amplitude of noise level, where $\kappa = 0$ presents determinate occupation, while ∞ indicates stochastic occupation. After each time step, each walker will change its strategy with a tunable parameter P_C ($0 \leqslant P_C \leqslant 1$) (i.e., a probability to change the present strategy). Overlapping on one site is prohibited.

Figure 1 shows the presentation of the motion of walker 1. It is seen that there are two types of walkers, labeled 1–5 on the sites (without empty volumes) in which the red ones indicate cooperators and the blue ones are the defectors. Since the door is downward, walker 1 will move randomly to one position of the three sites 3, 4, and 5. It is assumed that walker 1 could get rid of the constraints of the potential well to move to the site occupied by walker 5; then the site will be occupied by walker 1, while walker 5 has to be pushed onto the site previously occupied by walker 1 [see Fig. 1(b)].

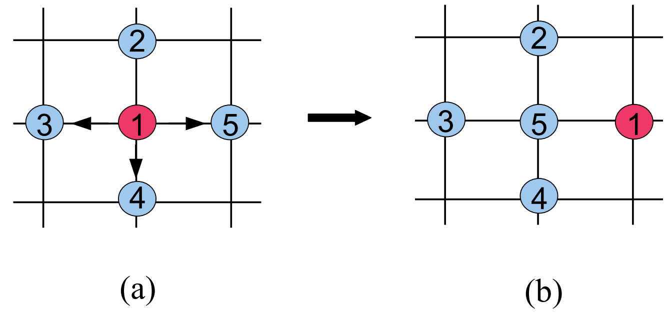

FIG. 1. (Color online) Schematic presentation of the motion of the walker 1.

III. SIMULATION AND ANALYSIS

A. Small-scale study

Simulations are carried out on a square lattice for a room scale 25×25 and the pedestrian scale $N = 500$. Initially, the pedestrians are randomly distributed in the room with the initial density of cooperators $\rho_{IC} = 0.5$, and each walker will change its strategy with a probability P_C ($0 \leqslant P_C \leqslant 1$) after each time step. In all of the simulations, each data point results from an average of 400 realizations. The room is set with only one door downward located at sites 12–14 with the length 3 sites and amplitude of noise level $\kappa = 0.1$.

Figure 2 shows the evolution of the number of pedestrians (N_P) in the room for different values of r (on the left), and the variation of average escape time with r at $P_C = 0.5$ (on the right). One can see on the left that small values of r (e.g., black solid squares at $r = 0.05$) lead to a long escape time, while larger values of r enable a shorter escape time (e.g., yellow solid right triangles for $r = 0.8$). Meanwhile (see Fig. 2 on the

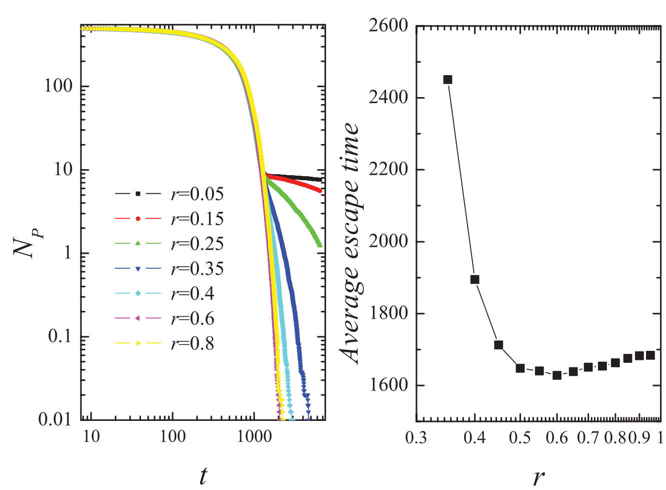

FIG. 2. (Color online) Left panel: Evolution of the number of pedestrians (N_P) in a house for different values of r at $\rho_{IC} = 0.5$, $P_C = 0.5$. Symbols for $r = 0.05$–0.8 correspond to black solid squares, red solid circles, green solid up triangles, blue solid down triangles, cyan solid diamonds, magenta solid left triangles, and yellow solid right triangles, respectively. Right panel: Average escape time as a function of r.

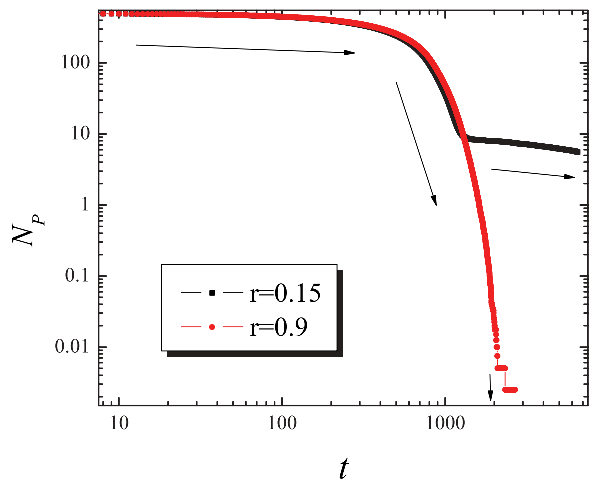

FIG. 3. (Color online) Evolution of the number of pedestrians N_P for $r = 0.15$ (black solid squares) and 0.9 (red solid circles), respectively. $\rho_{IC} = 0.5$, $P_C = 0.5$.

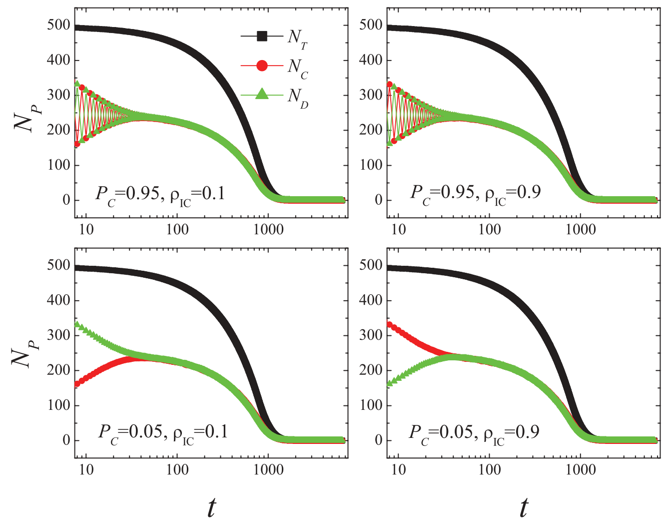

FIG. 4. (Color online) Variation of N_P with the average escape time at $P_C = 0.95$ and 0.05 for the initial density of cooperators $\rho_{IC} = 0.1$ and 0.9, respectively. N_T (black solid squares) is the total number of pedestrians in the room, N_C (red solid circles) is the number of cooperators, and N_D (green solid triangles) is the number of defectors among the pedestrians.

right), there exists a moderate value of $r = 0.6$ leading to the shortest escape time, and the situations of larger values of r are better than that for the small ones in general.

Precisely, evolutions of N_P for $r = 0.15$ and 0.9 are shown in Fig. 3, respectively. It is seen that when $r = 0.15$ (see the black symbols in Fig. 3), there are three phases during the whole process: In the beginning, the pedestrian flow moves at a small rate due to the large scale of pedestrians, and then remarkably speeds up as the number of walkers decreases to a value about $N_P = 200$, but finally moves at a very slow speed when N_P is very small (about 10). However, for $r = 0.9$, there are mainly two phases in the process: In the first stage, the trend of the pedestrian flow is the same as that for $r = 0.15$, and when the number of N_P decreases to about 200, the flow moves at a very large speed until all of the pedestrians escape from the room.

Before further analysis, it is necessary to discuss the results shown above and analyze the function of r in this model. We know that r influencing the values in the payoff matrix in Table I actually reflects the conflict between a cooperator and defector. Precisely, when r is small, there is little difference among T ($T = 1 + r$), R ($R = 1$), and S ($S = 1 - r$), which represents the small conflict between a cooperator and defector; while for large values of r, there are large differences between T, R, and S, indicating big conflicts between a cooperator and defector. And meanwhile, small conflicts induce the small diversities of payoff among the pedestrians, while large conflicts will lead to the large inhomogeneities of payoff. As a result, when r is small, there is little diversity between P_x and U_x according to the motion law [see Eq. (1)], which leads to a more consistent motion for each walker, especially when $r \to 0$, when the model returns to the biased random walks. However, for large values of r, big inhomogeneities between P_x and U_x lead to the inhomogeneous motions among the pedestrians, which induce a larger flow rate of pedestrians than that for small values of r. Based on the analysis, r plays a critical role in this model regarding influencing the flow rate of pedestrians. Actually, to be better understood, r can be realized as the fear index (VIX) of a walker: small values of r indicate an unconcerned attitude about the present situation, while large values of r represent the intense emotion of a pedestrian. It is shown in Fig. 2 that it is adverse for the people when they are too relaxed or too nervous, and keeping a moderately intense emotion is most favorable to prevent emergencies.

We further study the evolution of cooperators and defectors among the pedestrian flow in Fig. 4 for different values of P_C (i.e., the probability to change the present strategy) and ρ_{IC} (i.e., the initial density of cooperators) at $r = 0.5$. The black solid symbols are the presentation of total pedestrians (N_T), the red solid circles are the number of cooperators (N_C), and the green solid triangles are the number of defectors (N_D). It is shown that cooperators and defectors always arrive at a consistent state in which they have the same number and follow the same rule versus the evolving time, irrespective of

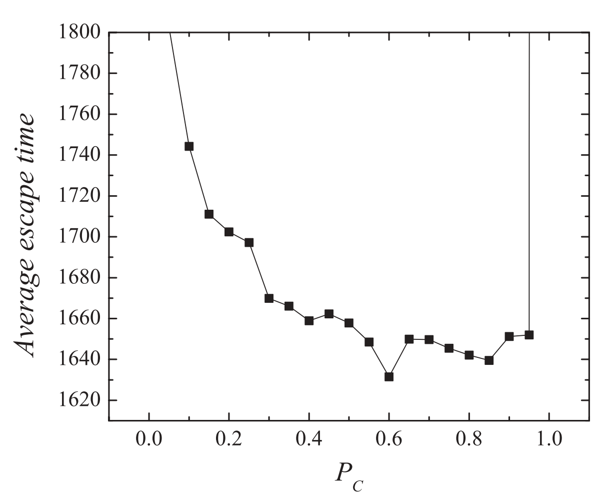

FIG. 5. Average escape time as a function of the probability of changing strategy P_C at $r = 0.5$, $\rho_{IC} = 0.5$.

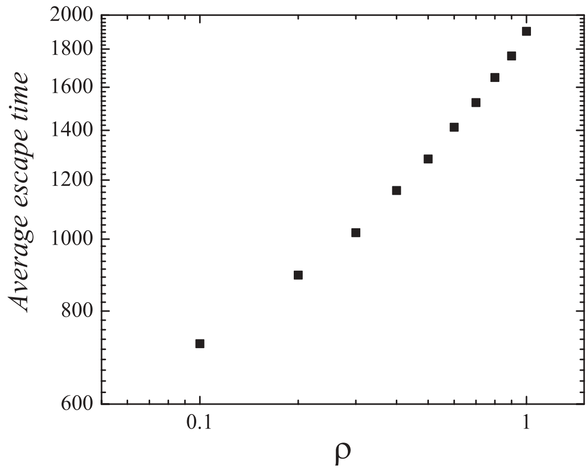

FIG. 6. Density of pedestrians ρ in a room as a function of the average escape time at $r = 0.5$, $\rho_{IC} = 0.5$, and $P_C = 0.5$.

ρ_{IC} and P_C. In addition, different dynamics are played before they reach the consistent state when P_C is different: for $P_C = 0.95$, the number of cooperators and defectors oscillates on the initial stage, while for $P_C = 0.05$, no oscillation happens. We also test the situations for different values of P_C and ρ_{IC}, and conclude that oscillation always exists when $P_C > 0.5$, while for $P_C \leqslant 0.5$, no oscillation emerges, irrespective of ρ_{IC}. The existence of a consistent state can be ascribed to the self-organization effect of the pedestrian flow. Excitedly, we also investigated the time point of the onset of the steady state for different scales of pedestrians and found that the time point is unrelated to the scale of pedestrians.

Based on the results in Fig. 4, the effect of P_C on the average escape time is investigated in Fig. 5. It is seen that it consumes the most time for $P_C = 0$ and 1, and, in general, larger values of P_C save the time remarkably well. That means that it is not favorable to escape from the room for a pedestrian with an attitude too stubborn or too capricious, and people with a flexible capability can minimize the escape time.

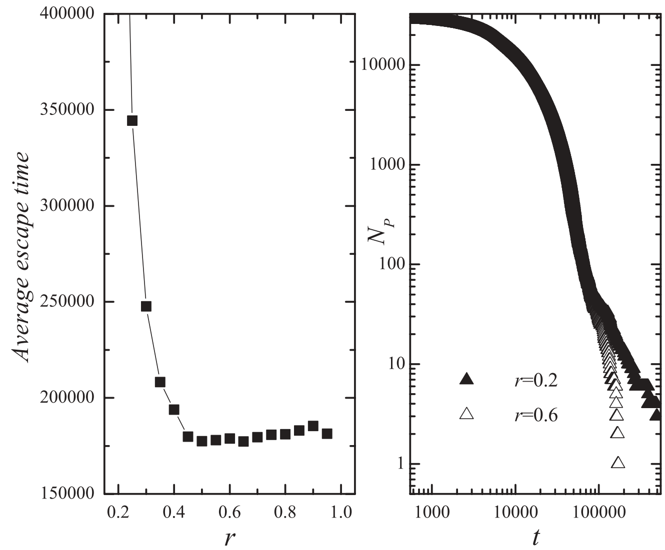

FIG. 8. Left panel: Average escape time as a function of r. Right panel: Evolution of the number of pedestrians (N_P) in a room for different values of r at $\rho_{IC} = 0.5$, $P_C = 0.5$.

Finally, variation of the pedestrian density ρ with the average escape time is shown in Fig. 6, and the evolution of pedestrians in the room is shown in Fig. 7. It is proven that the average escape time increases as ρ increases; that is, smaller density ρ induces a shorter average escape time.

B. Large-scale study

We have discussed the pedestrian evacuation in a room with the large scale 200×200 and pedestrian scale 30 000 using this model. It is shown that no large qualitative changes happen compared with that on the small scale of pedestrians.

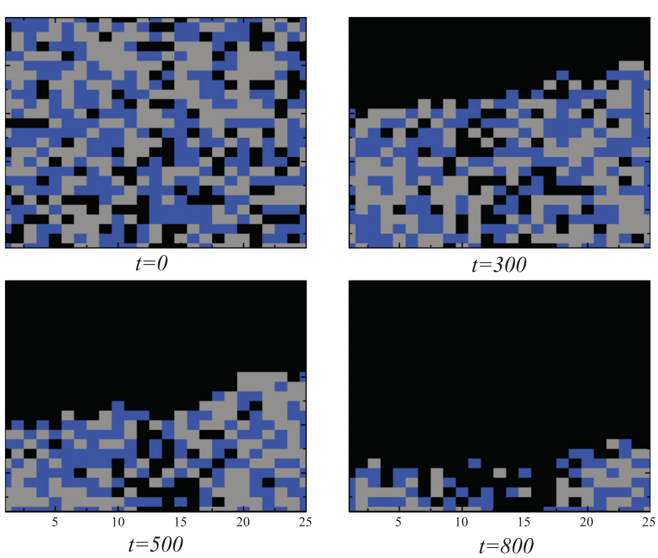

FIG. 7. (Color online) Evolution of pedestrians on the square lattice 25×25 in which the black symbols are empty sites, the brown ones are defectors, and the blue ones are cooperators.

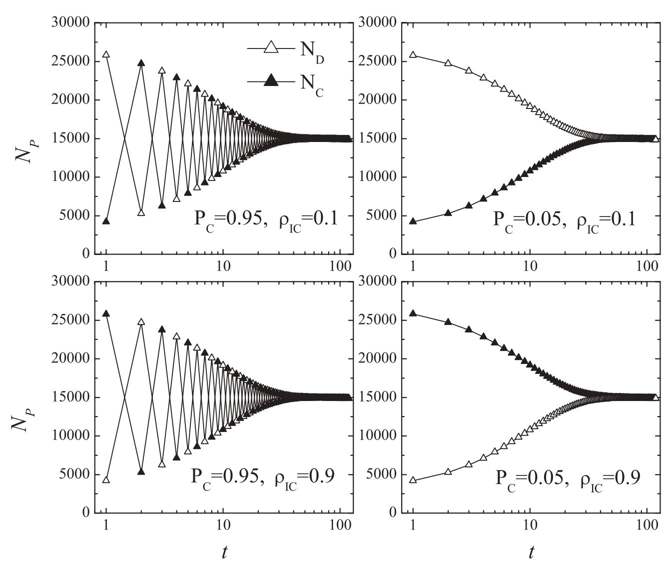

FIG. 9. Variation of N_P with the average escape time at $P_C = 0.95$ and 0.05 for initial density of cooperators $\rho_{IC} = 0.1$ and 0.9, respectively.

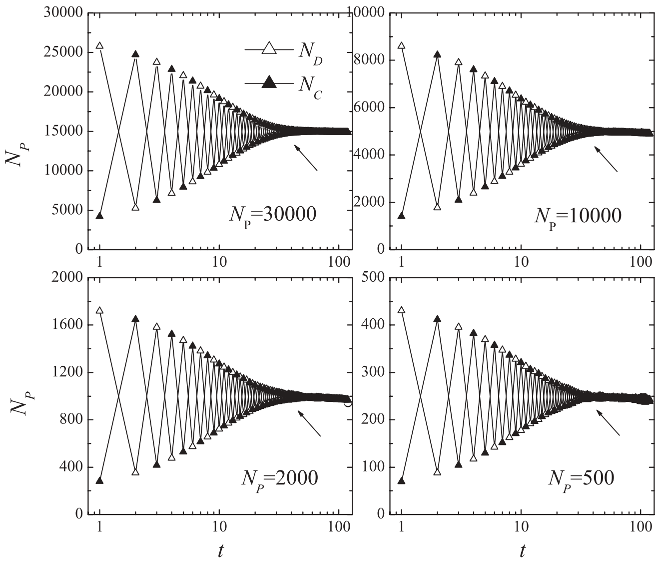

FIG. 10. Variation of N_P with the average escape time at $P_C = 0.95$ and $\rho_{IC} = 0.1$ for different scales of pedestrians, respectively.

Figure 8 (on the left) shows the variation of average escape time with the cost-to-benefit ratio r. It is seen that moderate values of r induce a shorter escape time, and the case for large values of r is better than small ones, in general. Meanwhile, three stages exist during the whole process (on the right in Fig. 8), but different dynamics are played on the last stage: when r is too small, the flow rate becomes smaller than before, while for larger values of r, the flow moves faster.

Note in Fig. 9 also that a consistent state exists: both the numbers of cooperators and defectors oscillate for $P_C > 0.5$, while no oscillation emerges when $P_C \leqslant 0.5$. Besides, the time point of the onset of the steady state is the same (about 32 steps), irrespective of the pedestrian scale shown in Fig. 10. Compared with the results in Fig. 4 on a small scale room of 625, we can conclude that the time point of the onset of the steady state is also unrelated to the scale of the square lattice.

P_C as a function of the average escape time is shown in Fig. 11, where one can see that the trend is consistent with that in Fig. 5: the most time is consumed for $P_C = 0$ and 1, and, in general, larger values of P_C save time remarkably well. That means that it is not favorable to escape from the room for a pedestrian with an attitude too stubborn or too capricious, and people with a flexible capability can minimize the escape time. These results coincide with the conclusions made in [33].

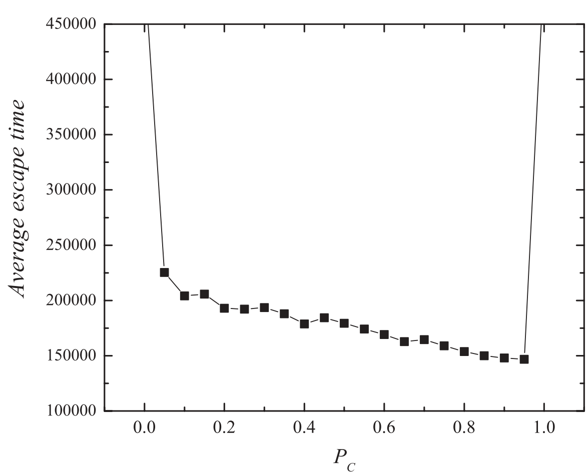

FIG. 11. Average escape time as a function of the probability of changing strategy P_C at $r = 0.5$, $\rho_{IC} = 0.5$.

IV. CONCLUSIONS

In this work, the evacuation of pedestrians was studied based on the modified lattice gas model by using the snowdrift game theory. The payoff matrix of the snowdrift game model represents the complicated interactions among the conflict pedestrians. It is found that each walker interacts with all of its neighbors, and chooses a neighbor randomly to occupy the site with a probability (motion law) $W(x \to y) = \frac{1}{1+\exp[-(P_x - U_x)/\kappa]}$, where P_x is the x's payoff representing his personal energy, and U_x is the average payoff of its neighborhood indicating the potential well energy if he stays. We studied the impacts of the cost-to-benefit ratio r on average escape time, and found that the situation is better, in general, for the case of large values of r, and an intermediate value of r induces the shortest escape time. Meanwhile, it is also seen that there are three phases during the whole process for small values of r, while two phases exist for large ones. We explained the results by analyzing the function of r, and concluded that r reflecting the conflicts among the walkers actually affects the flow rate of pedestrians: small values of r induce a more consistent motion for each walker, in particular, when $r \to 0$, and the model returns to the birandom walks; while for large values of r, inhomogeneous motions among the walkers lead to a larger flow rate than that for the small ones. So r can be understood as the fear index (VIX) of pedestrians: it is adverse for the people when they are too relaxed or too nervous, and keeping a moderate intense emotion is most favorable to eliminate emergencies. Moreover, the dynamics of the two types of walkers was studied, and it was shown that there exists a consistent state in the pedestrian flow system in which cooperators and defectors have the same number and evolve by the same law. In addition, it exhibits a different dynamics before the system reaches the consistent state: both the numbers of cooperators and defectors oscillate for $P_C > 0.5$, while no oscillation emerges when $P_C \leqslant 0.5$. We ascribe this phenomenon to the self-organization effect of pedestrians. Furthermore, we investigated the effects of pedestrian scale on the time point of the onset of the steady state, and proved that the time point is unrelated to the scale of pedestrians, as well as the scale of the square lattice. Finally, the effect of the pedestrian density ρ on the average escape time was discussed, and it was shown that the escape time decreases as ρ decreases.

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